QM1 statistics: week 6

Hypothesis Tests and Intervals for Means

Welcome!

A bloody affair

Hypothesis tests

Donating blood is always a good cause

- But did you know that some blood banks… are run by vampires that drink most of what gets donated?
  - Journalist Stoker suspected that vampires controlled more than 3% of all blood banks in the world
  - So he visited a random sample of $n = 500$ facilities
- If $\hat{p} > 0.03$… what would that prove?
  Sample proportions can be inaccurate, after all
- **Hypothesis tests** (also known as **significance tests**) aim to rise above the uncertainties of sampling

I State hypotheses

- A hypothesis test needs a starting point, a so-called **null hypothesis**:

$$H_0: p = 0.03$$

- This one basically says: *nothing is going on*
  (it is indeed the case that vampires run 'only' 3% of all blood banks)
- Stoker's aim was to prove this null hypothesis wrong. If he succeeded, he would demonstrate one of three possible **alternative hypotheses**:

$$\boxed{\begin{array}{c} H_A: p < 0.03 \\ H_A: p > 0.03 \end{array}}$$

One-sided test:
for when you have a certain expectation
(be a bit careful with this)

or

$$\boxed{H_A: p \neq 0.03}$$

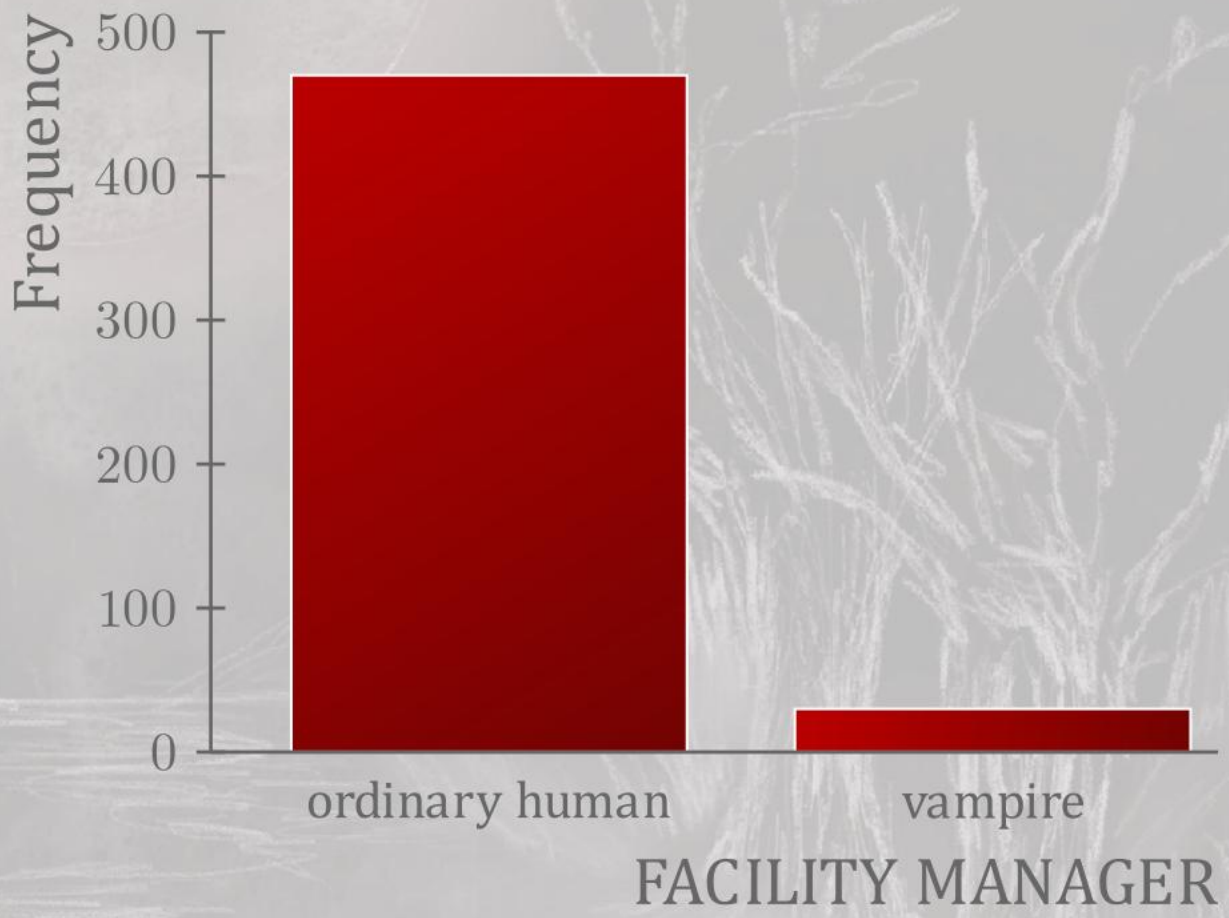
Two-sided test
(most common)

II Draw the sample

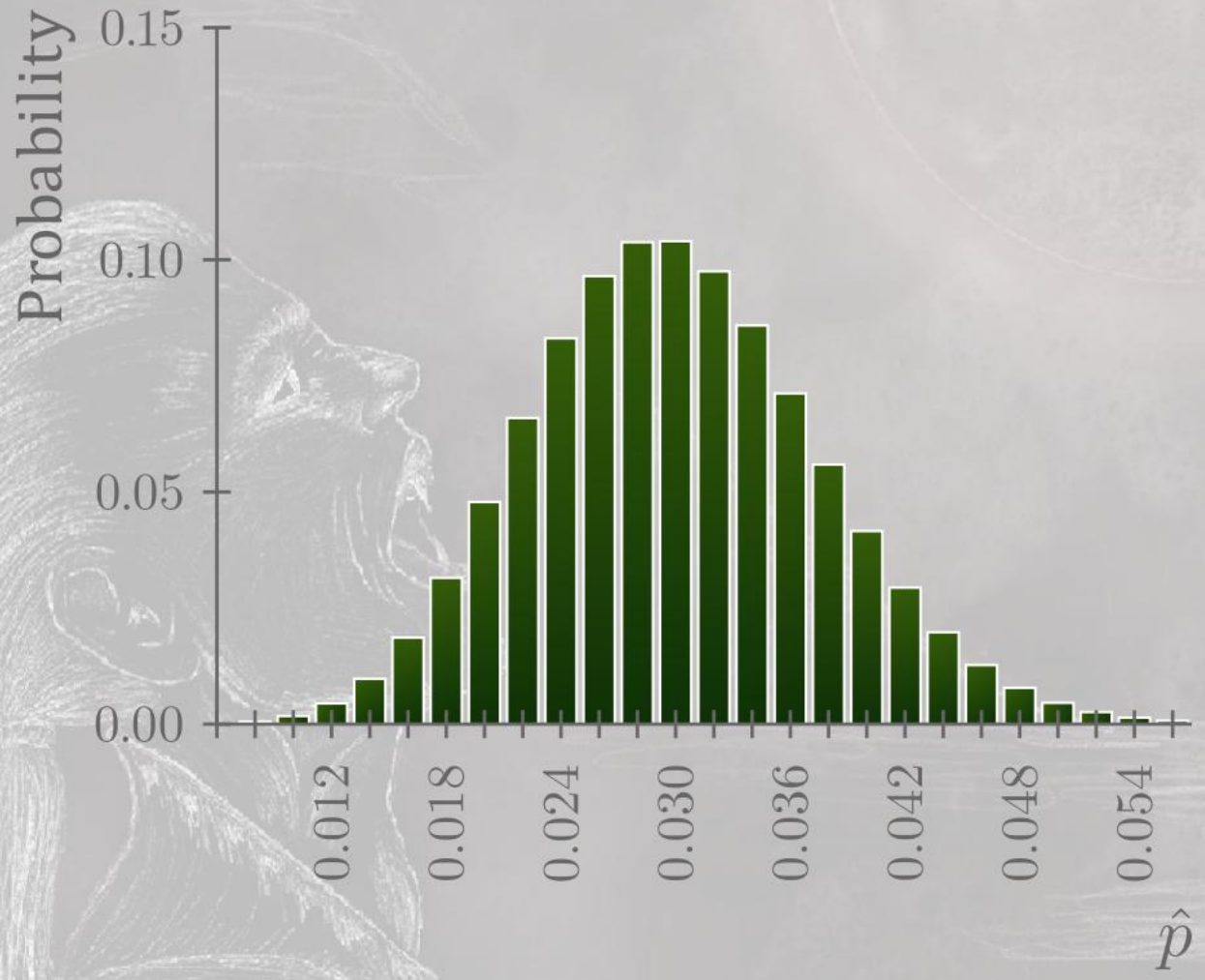
FACILITY MANAGER	ordinary human	470
	vampire	30
	TOTAL	**500**

$$\hat{p} = \frac{x}{n} = \frac{30}{500} = 0.06$$

- Twice as much as 0.03… but that might be due to *chance*!
- Perhaps this sample just overestimates the population proportion?
- What's the *probability* of such an overestimation?



III Calculate a probability

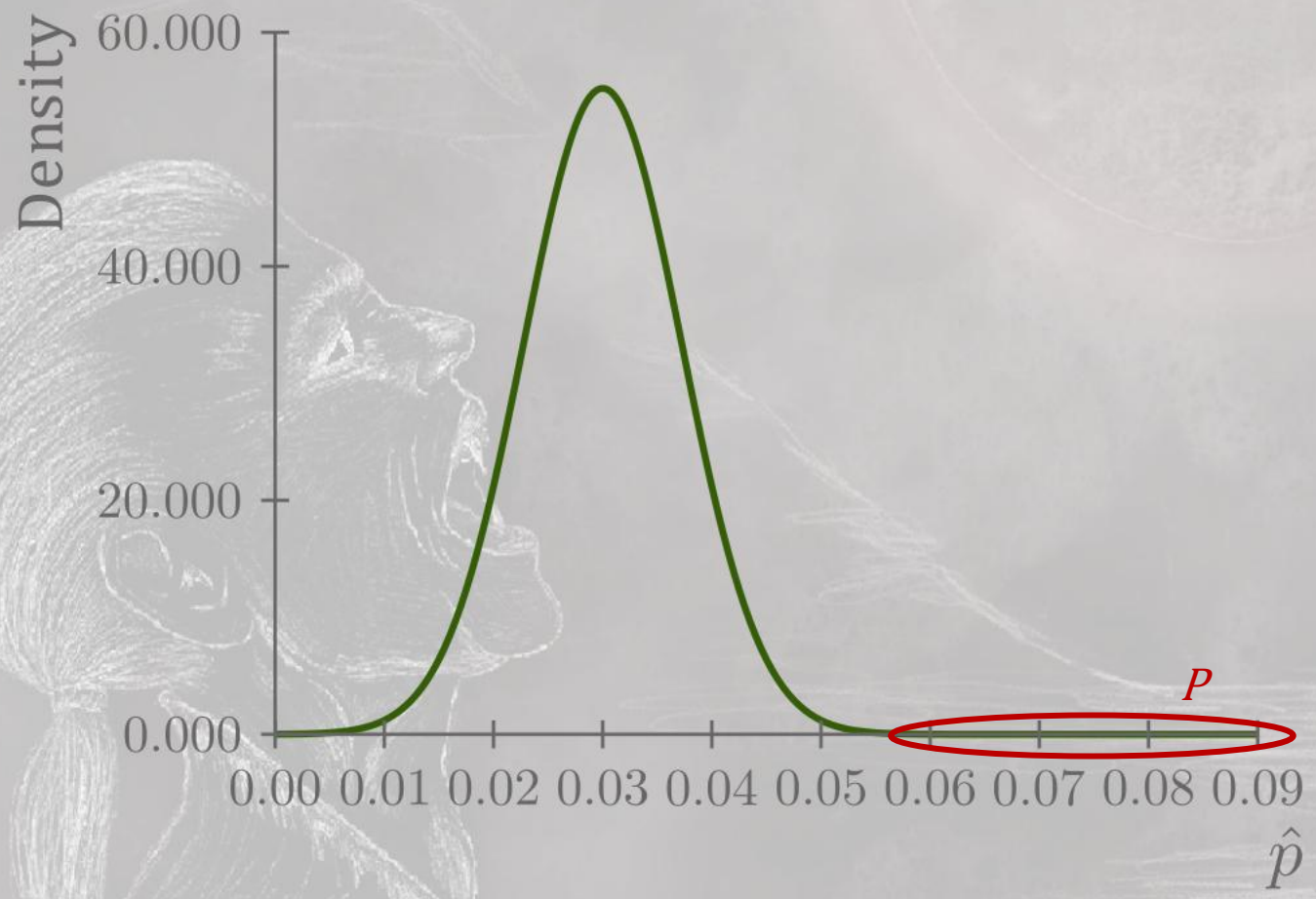


- For this, we have to use the **sampling distribution** of $\hat{p}$
- Let's come up with the parameters:
  - If the null hypothesis is true,
    $$\boldsymbol{\mu(\hat{p}) = p_0} = 0.03$$
  - Also,

$$\boldsymbol{\sigma(\hat{p})} = \sqrt{\frac{\boldsymbol{p_0 q_0}}{\boldsymbol{n}}} = \sqrt{\frac{0.03 * 0.97}{500}}$$
$$= 0.0076$$

- The shape is approximately normal, thanks to the Success/Failure condition ☺
  - $np = 500 * 0.03 = 15$
  - $nq = 500 * 0.97 = 485$

III Calculate a probability (continued)



- Again: if the population proportion was 0.03, how often would we overestimate it by at least twice that amount)? How big is

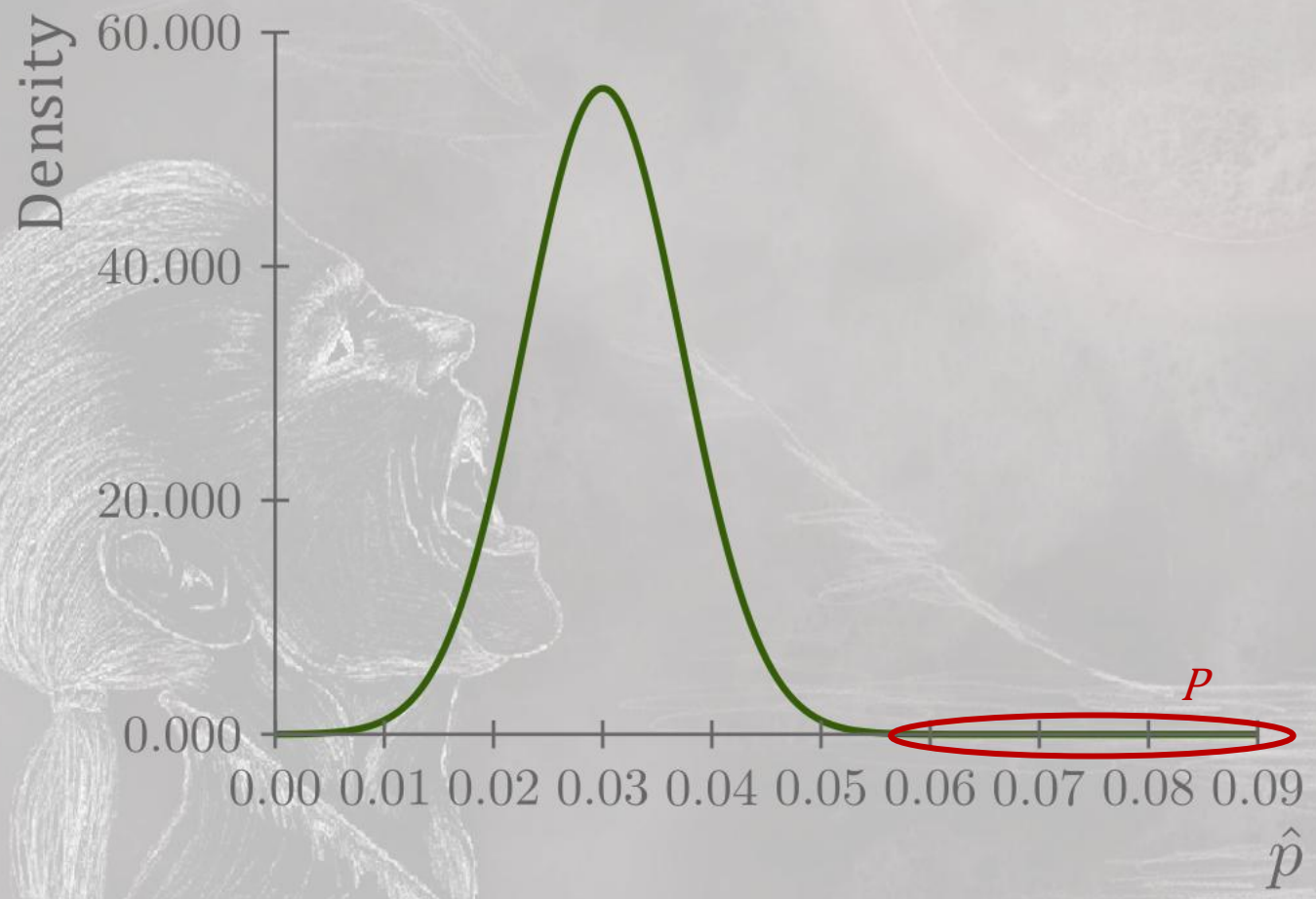
$$P(\hat{p} \geq 0.06 | p = 0.03)?$$

ARGH! WTF is that notation? ☹

- P: probability
- $\hat{p}$: sample proportion
- p: population proportion

- Should this event be pretty probable: no reason to doubt the null hypothesis
- Should this event be very improbable: had Stoker then truly visited such an unlikely sample of facilities, or was the population proportion simply larger than 0.03?

III Calculate a probability (continued)



- Let's now determine it
- First, we convert $\hat{p}$ to a *z-score*:

$$z = \frac{x - \mu}{\sigma} = \frac{\hat{p} - p_0}{\sigma(\hat{p})} =$$

$$\frac{0.06 - 0.03}{0.0076} = 3.93$$

- Now get the z-table (see the last lecture)
- It turns out that

$$P(z \geq 3.93) < 0.0001$$

- This is the **p-value**
  - The probability that you would obtain a sample result like this one – or an even more extreme result – *if the null hypothesis were true*

IV Decide

- 'What? The probability that the null hypothesis is true is less than 0.01%?'
- No, that's not what I said ☺
- Stoker's p-value indicates: *if the null hypothesis were true*, it would be extremely unlikely for him to draw a sample of blood banks where at least 6% (0.06) are run by vampires
  - When n = 500, this happens less than 1 in 10,000 times
- Such a tiny probability suggests that the null hypothesis was actually *not* true (since Stoker *had* found that sample proportion!)
- So a small p-value allows you to *reject* the null hypothesis: probably, *more than 3%* of all blood banks (the population) have been taken over

IV Decide (continued)

- But what kind of p-value would be small enough?
- Let's agree on a threshold value beforehand: the **significance level α**
  - Commonly, $\alpha = 0.05$
- Is your p-value smaller than α? Then your test result is *significant*: it 'signifies' that something may be wrong with the null hypothesis
- Is your p-value larger than α? Then you cannot reject the null hypothesis
  - Careful: the latter does *not* need to mean that the null hypothesis is true! Only that you *don't have enough evidence against it*

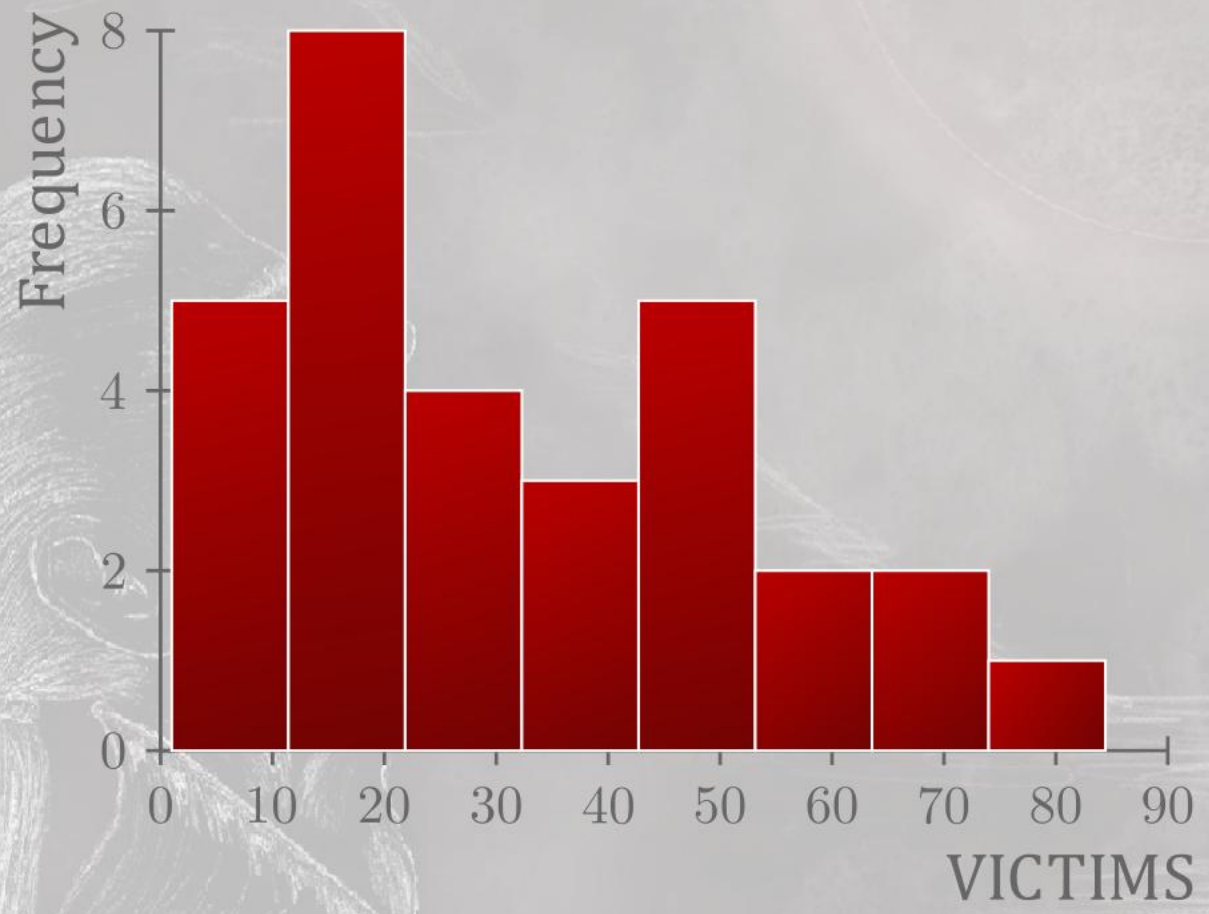
In summary: the action plan

1. State the hypotheses
2. Choose the significance level (commonly 0.05)
3. Draw the sample and calculate the p-value: $P(data|H_0)$
4. Compare the p-value with the significance level, and cast your judgment on the null hypothesis

A century old

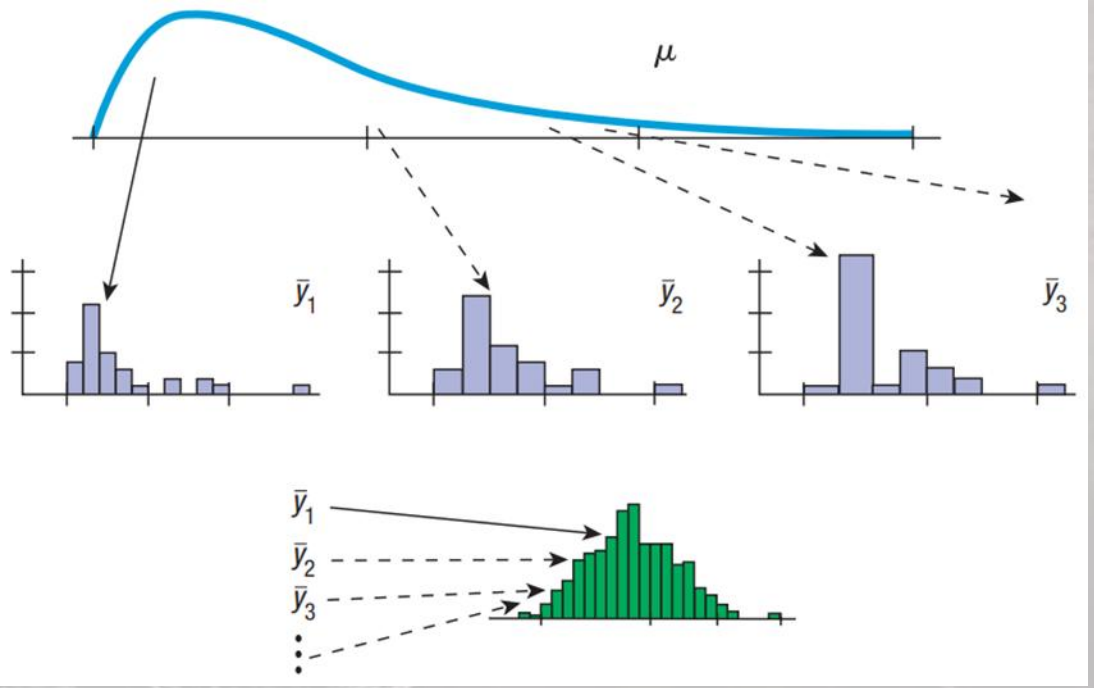
Confidence intervals for means

It turns out that Stoker had a double agenda…



- He approached the 30 vampires who owned a blood bank, and offered them a course in contemporary fashion
  - This intervention might help them to disguise themselves as a common citizen, since many had a habit of wearing black tuxedos and long black robes with extremely high collars
- Once a vampire completed the training, Stoker asked them to record how many blood donors they got during the next 2 months
  - This is the variable of interest: the number of unsuspecting victims. The higher, the better
  - It's a <u>quantitative</u> variable!
- Here are the sample results. $\bar{y} = 32.2$ (and $s = 20.2$)
- But what could be the true population mean?

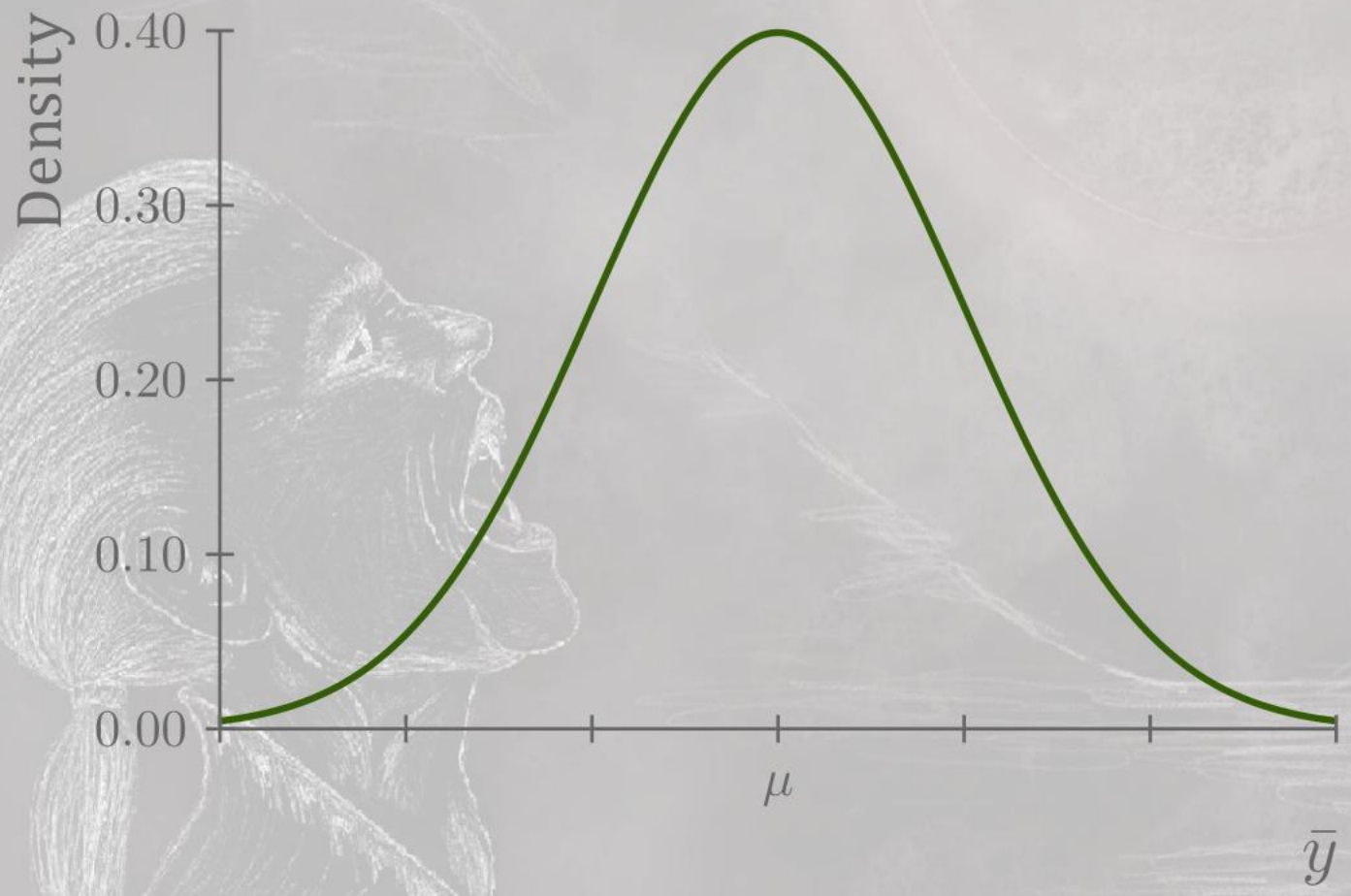
This sample mean (32.2) is just one of the many we could have obtained



Picture adapted from Sharpe, figure 11.6 (page 351)

- Stoker was actually interested in the **population distribution** (the victim counts of all vampires who disguise themselves well). The parameter he wanted is:
  - μ (the population mean)
- But he only drew a random sample of 30 vampires, which provided him with a **distribution of sample scores** (displayed on the previous slide). It yielded a statistic/estimator:
  - $\bar{y}$ (the sample mean)
- And this random sample is just one of the many that Stoker could have drawn. As such, the sample mean is a **random variable**
- The **sampling distribution of $\bar{y}$** describes all the values that $\bar{y}$ could take on, and how likely each value is. Important parameters:
  - $\mu(\bar{y})$ (the expected value of the sample mean)
  - $\sigma(\bar{y})$ (the standard deviation of the sample mean)

The sampling distribution of $\bar{y}$: key characteristics

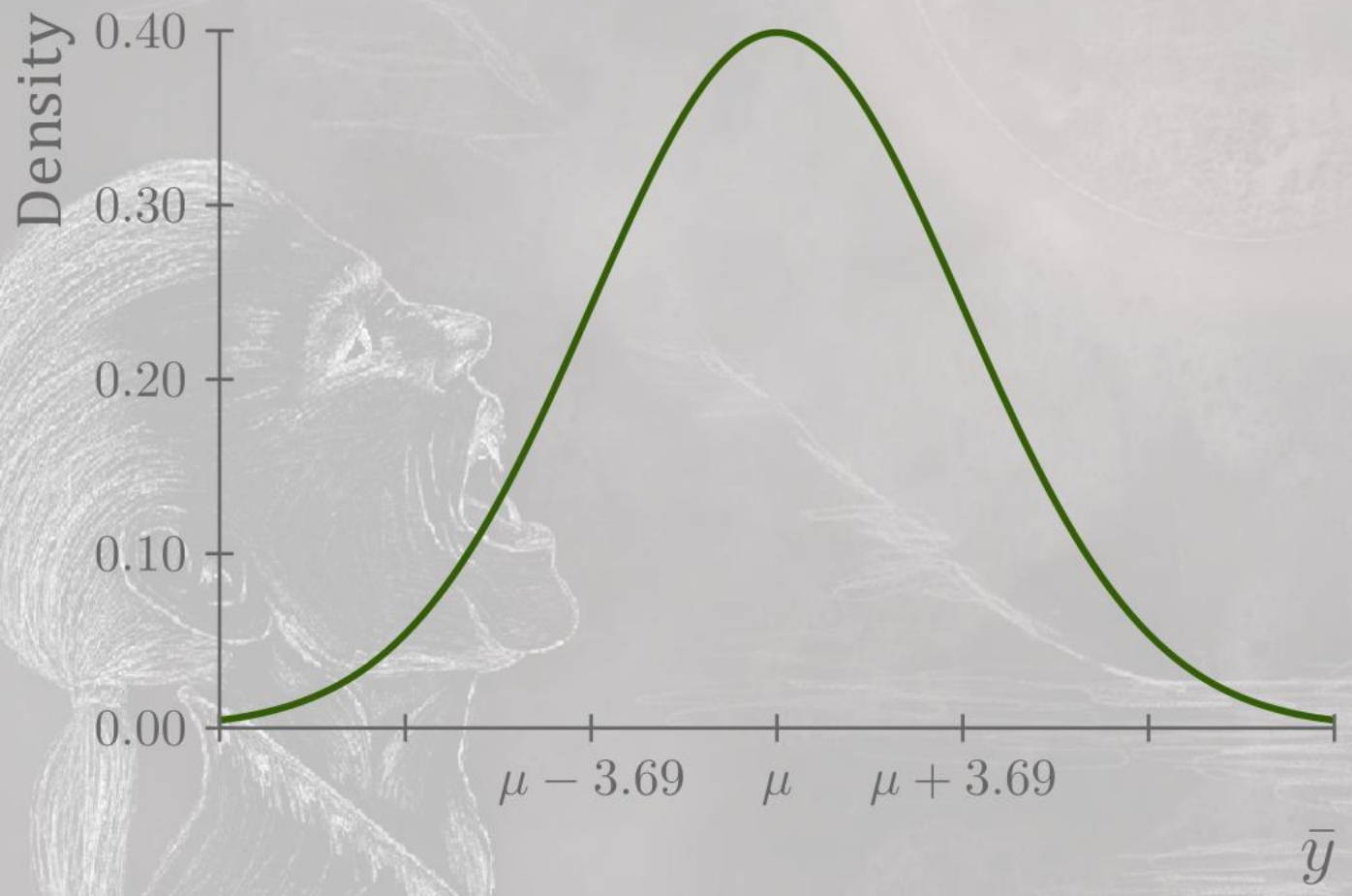


- As long as *n* is large enough, its shape will be approximately **normal** ☺
  - This rule is actually just a more general case of the Success/Failure condition (for binomial distributions)
  - We call it the **central limit theorem** (more information in the textbook)
- Its <u>expected value</u> will be the true population mean μ

$$\mu(\bar{y}) = \mu$$

  - So: the <u>average</u> sample would get the population mean exactly right
  - Think back to the binomial distribution where $\mu(\hat{p}) = p$
- How about the <u>standard deviation</u>?
  - How much will the sample mean usually deviate from the <u>true</u> average number of victims per vampire (μ)?

The standard deviation / standard error of $\bar{y}$



- You can calculate the standard deviation of the sample mean like this:

$$\sigma(\bar{y}) = \frac{\sigma}{\sqrt{n}}$$

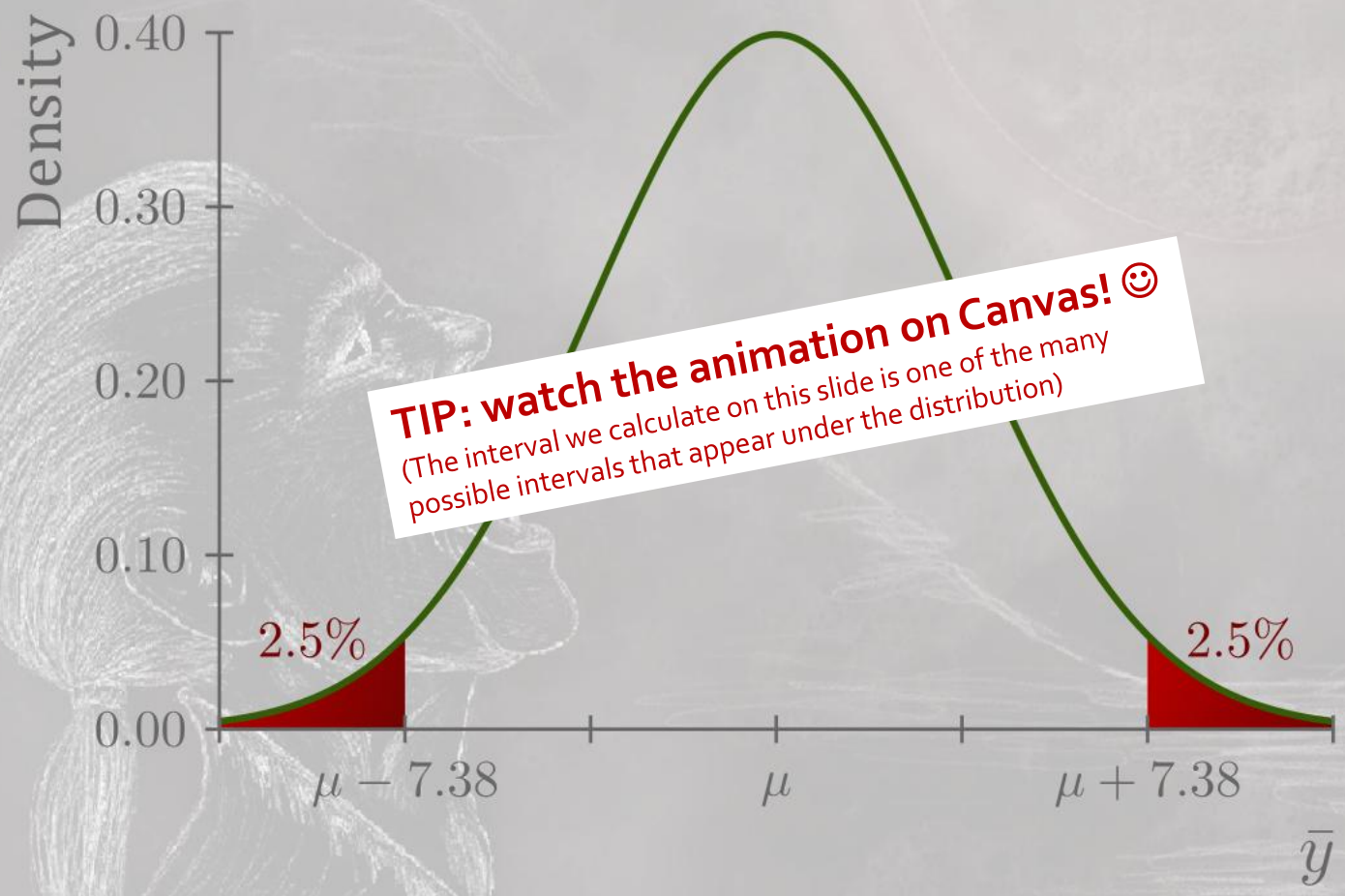
- σ is the standard deviation of the <u>population</u> distribution (which describes the spread among <u>individual</u> vampires), and n is the sample size
- Example: if $\sigma = 20$ and $n = 30$,

$$\sigma(\bar{y}) = \frac{20}{\sqrt{30}} = 3.65$$

  - On average, the sample means you can draw will over-/underestimate the population mean by 3.65 donors
- But actually, we don't know σ. Shall we just estimate it with s then (the standard deviation of the sample)?
- We call this estimate the **standard error**:

$$SE(\bar{y}) = \frac{s}{\sqrt{n}} = \frac{20.2}{\sqrt{30}} = 3.69$$

Now we can make a confidence interval again! ☺



TIP: watch the animation on Canvas! ☺
(The interval we calculate on this slide is one of the many possible intervals that appear under the distribution)

- Remember how we did this with proportions (categorical variable):

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

sample estimate — margin of error

- We can apply the same principle with means (quantitative variable):

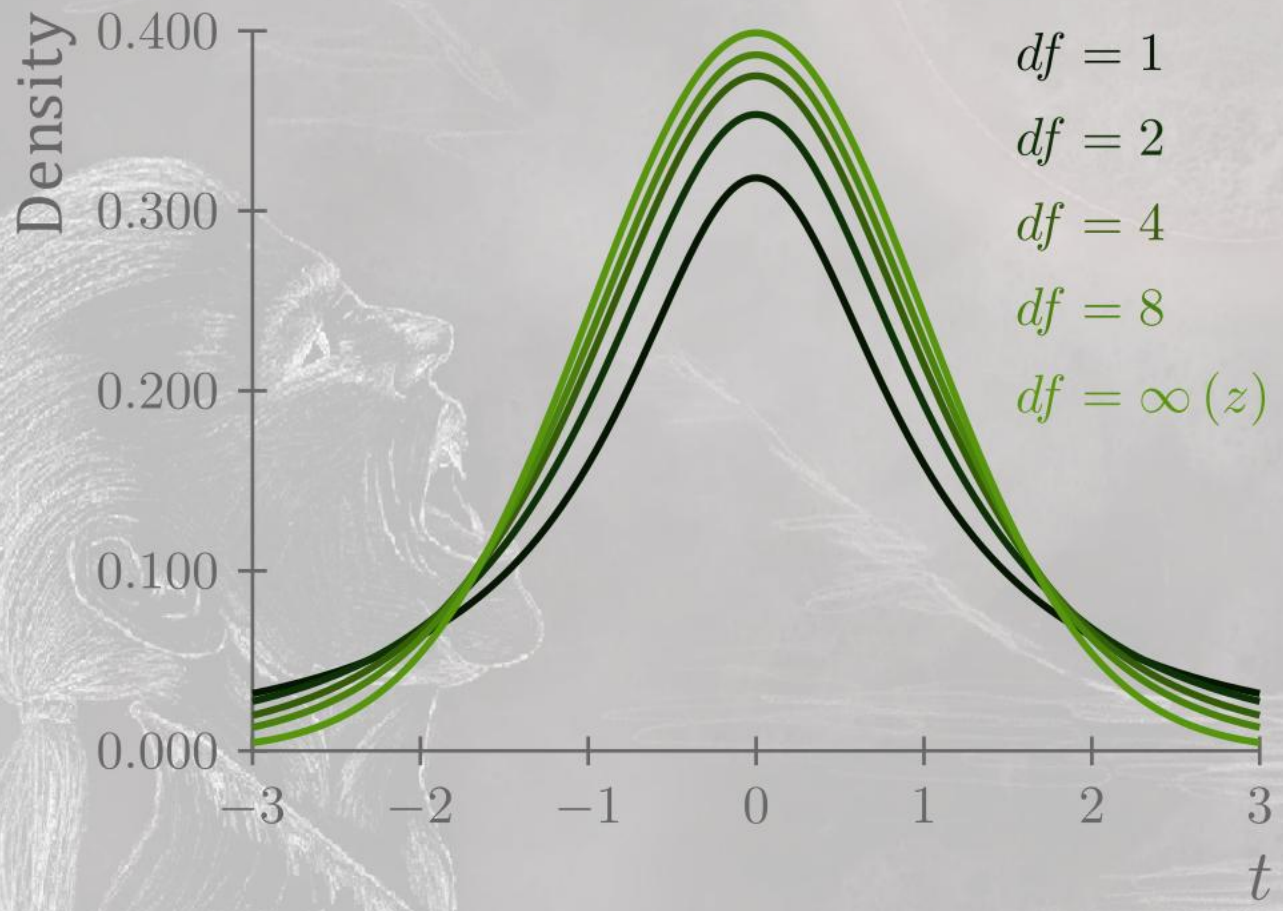
$$\bar{y} \pm z^* \frac{\sigma}{\sqrt{n}}$$

- For a 95% confidence interval, we use $z^* = 1.96$ or $z \approx 2$. In that case,

$$32.2 \pm 2 * \frac{20.2}{\sqrt{30}}$$
$$32.2 \pm 7.38$$
$$[24.82, 39.58]$$

- But hang on… something about this calculation isn't quite right

We are estimating σ.
That creates more uncertainty



- When we use s in our calculation and not σ, we should not work with a normal distribution (a.k.a. z-distribution)
- After all, the standard error of $\bar{y}$ probably isn't 100% accurate

$$\frac{s}{\sqrt{n}} \neq \frac{\sigma}{\sqrt{n}}$$

- So let's work with a so-called **t-distribution**
  - The t-distribution is flatter and more spread out
  - But the shape of the t-distribution depends on the sample size. As *n* grows, t will get closer to z again
  - More exactly, the shape depends on the so-called **degrees of freedom**:

$$df = n - 1$$

  - 'What are those?' Technical detail. All you need to know for now is that, with $n = 30$, we have

$$df = 29$$

An improved 95% confidence interval

- The z-formula was:

$$\bar{y} \pm z^* \frac{\sigma}{\sqrt{n}}$$

- The t-formula becomes:

$$\bar{y} \pm t^* \frac{s}{\sqrt{n}}$$

$$32.2 \pm 2.045 * \frac{20.2}{\sqrt{30}}$$

$$32.2 \pm 7.54$$

$$[24.66, 39.74]$$

- TL;DR: with 95% confidence, Stoker could conclude: the weekly number of humans that the average vampire was able to fool into donating blood lay between 24.66 and 39.74 victims per month

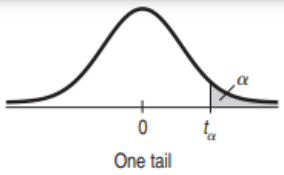


table T in Sharpe

degrees of freedom

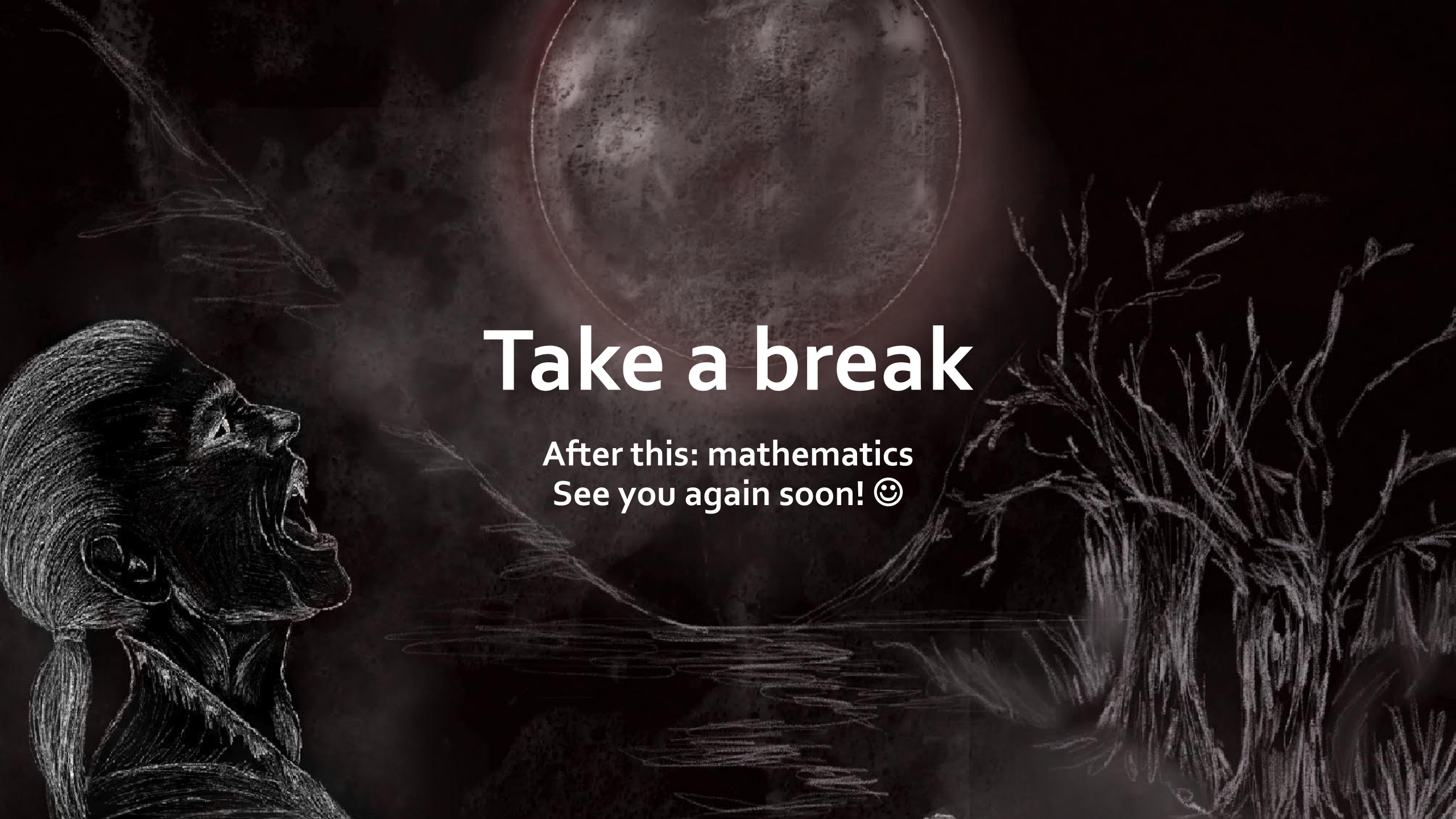
12	1.356	1.782	2.179	2.681	3.055	12
13	1.350	1.771	2.160	2.650	3.012	13
14	1.345	1.761	2.145	2.624	2.977	14
15	1.341	1.753	2.131	2.602	2.947	15
16	1.337	1.746	2.120	2.583	2.921	16
17	1.333	1.740	2.110	2.567	2.898	17
18	1.330	1.734	2.101	2.552	2.878	18
19	1.328	1.729	2.093	2.539	2.861	19
20	1.325	1.725	2.086	2.528	2.845	20
21	1.323	1.721	2.080	2.518	2.831	21
22	1.321	1.717	2.074	2.508	2.819	22
23	1.319	1.714	2.069	2.500	2.807	23
24	1.318	1.711	2.064	2.492	2.797	24
25	1.316	1.708	2.060	2.485	2.787	25
26	1.315	1.706	2.056	2.479	2.779	26
27	1.314	1.703	2.052	2.473	2.771	27
28	1.313	1.701	2.048	2.467	2.763	28
29	1.311	1.699	2.045	2.462	2.756	29
30	1.310	1.697	2.042	2.457	2.750	30
32	1.309	1.694	2.037	2.449	2.738	32
35	1.306	1.690	2.030	2.438	2.725	35
40	1.303	1.684	2.021	2.423	2.704	40
45	1.301	1.679	2.014	2.412	2.690	45
50	1.299	1.676	2.009	2.403	2.678	50
60	1.296	1.671	2.000	2.390	2.660	60
75	1.293	1.665	1.992	2.377	2.643	75
100	1.290	1.660	1.984	2.364	2.626	100
120	1.289	1.658	1.980	2.358	2.617	120
140	1.288	1.656	1.977	2.353	2.611	140
180	1.286	1.653	1.973	2.347	2.603	180
250	1.285	1.651	1.969	2.341	2.596	250
400	1.284	1.649	1.966	2.336	2.588	400
1000	1.282	1.646	1.962	2.330	2.581	1000
∞	1.282	1.645	1.960	2.326	2.576	∞
Confidence levels	80%	90%	95%	98%	99%	

t^*

Further topics for this week (self-study ☺)

- Sharpe, chapter 11
  - The central limit theorem
  - Assumptions and conditions
- Sharpe, chapter 12
  - Hypothesis tests for a mean (but: next week's lecture will discuss this as well)
  - Using confidence intervals as hypothesis tests

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture



Take a break

After this: mathematics
See you again soon! ☺